Unit 2 FRQ Set 1

Forces & Translational Dynamics · 4 free-response questions in real AP exam order

40 points Time: **95 min** (the AP exam's FRQ section) MR 10 · TBR 12 · LAB 10 · QQT 8

- Read all four questions first (2 min). Start with the one you're most sure of.
- **Derive** = start from a law in symbols. **Calculate** = equation, numbers in, units out. **Justify** = claim → law → link to THIS situation.
- Never leave a part blank. Points are awarded line by line.
- When the timer ends, stop. Grade yourself with the coaching guide's rubric, and be strict.

Reference (same ideas as the AP equation sheet)

$\Sigma F = ma$ · $F_g = mg$

$f_s \le \mu_s F_N$ · $f_k = \mu_k F_N$

$x_{cm} = \Sigma m_i x_i / \Sigma m_i$

$F_g = Gm_1m_2/r^2$ · $g = GM/r^2$

$a_c = v^2/r$ · $v = 2\pi r/T$

$F_s = -k\Delta x$

$v = v_0 + at$

$\Delta x = v_0 t + \frac{1}{2}at^2$

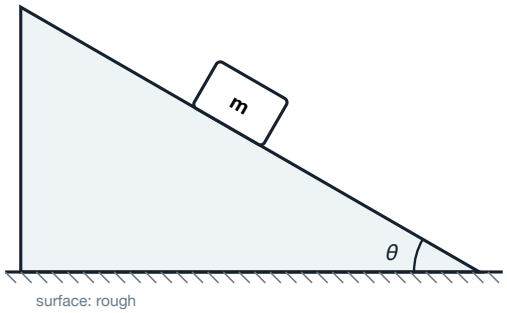
$v^2 = v_0^2 + 2a\Delta x$

$g = 9.8\ m/s^2$ · $G = 6.67 \times 10^{-11}\ N \cdot m^2/kg^2$

Question 1: Mathematical Routines (MR)

10 points · suggested 20–25 min

A block of mass m is released from rest at the top of a ramp of length D that makes an angle θ with the horizontal. The coefficient of kinetic friction between the block and the ramp is μ_k.



A. On the dot below, which represents the block, draw and label the forces (not components) exerted on the block as it slides down the ramp. Each force must be represented by a distinct arrow starting on, and pointing away from, the dot. (2 pts)

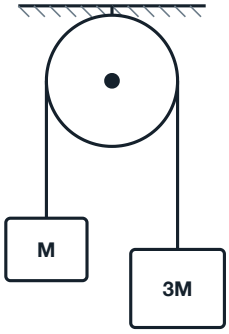
B. Derive an expression for the magnitude of the block's acceleration down the ramp. Express your answer in terms of θ, μ_k, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information. (3 pts)

C. For $\theta = 30°$, $\mu_k = 0.20$ and D = 2.0 m, calculate (i) the acceleration and (ii) the time for the block to reach the bottom of the ramp. `(2 pts)`

D. A student claims: “A block with twice the mass, made of the same material, would reach the bottom in less time, because the heavier block has a bigger force pulling it down the ramp.” Do you agree or disagree with the student? Justify your answer. `(3 pts)`

Question 2: Translation Between Representations (TBR) 12 points · suggested 25–30 min

Block 1 (mass M) and Block 2 (mass 3M) hang from opposite ends of a light string that passes over an ideal (massless, frictionless) pulley. The blocks are released from rest.



A. On the two dots below, draw and label the forces exerted on Block 1 and on Block 2 after release. The relative lengths of the arrows should reflect the relative magnitudes of the forces, both within each diagram and between the two diagrams. (3 pts)

Block 1 Block 2

B. Derive expressions for (i) the magnitude of the acceleration of the blocks and (ii) the tension in the string. Express your answers in terms of M and physical constants, as appropriate. (4 pts)

C. On the axes below, sketch the velocity of Block 2 as a function of time from release until just before it hits the floor (take downward as positive). On the same axes, sketch with a dashed line the velocity of an object that is dropped from rest in free fall. `(3 pts)`

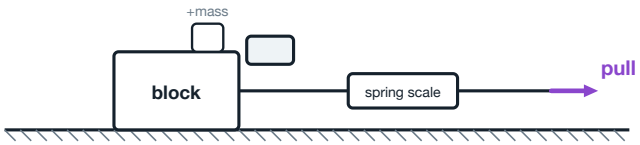


D. Is your result for the tension in part B consistent with your free-body diagrams in part A? Justify your answer. `(2 pts)`

Question 3: Experimental Design and Analysis (LAB)

10 points · suggested 25–30 min

Students want to determine the coefficient of kinetic friction μ_k between a wooden block and a lab table. They have the block, a set of masses that can be stacked on top of it, a spring scale, a balance, a meterstick and a stopwatch.

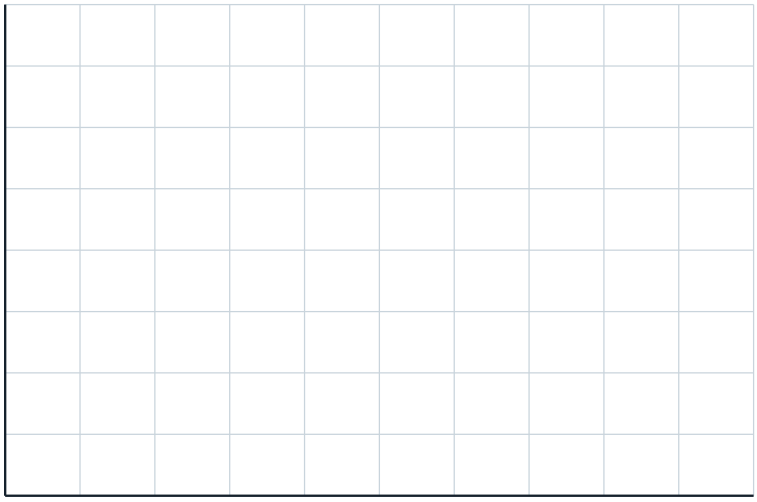


A. Describe an experimental procedure the students could use to collect the data needed to determine μ_k, using a graph. Include what is measured, how each measurement is taken, and how the students would reduce experimental uncertainty. (2 pts)

B. The students' data are shown below. Complete the table by filling in the normal-force column. Then indicate which quantities should be graphed to produce a straight line whose slope can be used to determine μ_k. (2 pts)

Total mass (kg)	Spring scale reading at constant speed (N)	Normal force (N)
0.50	1.6	
1.00	3.0	
1.50	4.6	
2.00	5.8	
2.50	7.5	

C. On the grid below, plot the data from part B. Clearly label the axes, including units, and use an appropriate scale. Draw a best-fit line for the data. `(4 pts)`

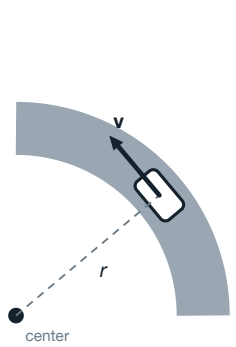


D. Using the best-fit line, calculate μ_k. `(2 pts)`

Question 4: Qualitative/Quantitative Translation (QQT)

8 points · suggested 15–20 min

Two identical cars drive around the same flat, unbanked curve of radius R. Car 2 travels at twice the speed of Car 1. Both cars make the turn without skidding.



A. WITHOUT using equations, explain how the minimum coefficient of static friction needed for Car 2 to make the turn compares with the minimum needed for Car 1. (3 pts)

B. Derive an expression for the minimum coefficient of static friction μ_{min} needed for a car moving at speed v to make the turn. Express your answer in terms of v, R and physical constants, as appropriate. (3 pts)

C. Explain how your expression in part B supports your reasoning in part A. By what factor does μ_{min} change? (2 pts)